Review Article

Global Neurocognitive Vulnerability in the Anthropocene: A Narrative Review of Physiological-Psychological Impacts of Extreme Heat and Evidence-Based Policy Interventions

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ABSTRACT

The Anthropocene is characterized by an unprecedented rise in extreme heat events, posing a systemic threat to global neurocognitive health. While physiological impacts are well documented, neurocognitive vulnerabilities, ranging from acute cognitive inefficiency to heatstroke-related neuroinflammation and psychiatric morbidity, remain inadequately integrated into public health policy. The purpose of this narrative review is to synthesize multidisciplinary evidence on the physiological and psychological pathways through which extreme heat may affect cognition and mental health, and to consider the implications for heat-health policy and nature-based interventions. The review draws on targeted searches of PubMed and Google Scholar and integrates evidence across evolutionary perspectives, molecular biology, pediatrics, psychiatry, occupational health, and urban planning. Evidence indicates that acute heat exposure can impair aspects of attention, executive function, and cognitive control, although the magnitude and direction of effects vary by task, exposure intensity, acclimatization, and individual strain. Severe heat illness can involve blood-brain barrier disruption and neuroinflammatory responses, while nocturnal heat can disturb thermoregulation and sleep. Emerging human evidence also links prenatal heat exposure with neurodevelopmental outcomes, but this literature remains limited and should be interpreted cautiously. Current heat-health planning gives greater attention to physical morbidity than to mental health and neurocognitive risks. Nature-Based Solutions (NbS) may contribute to heat mitigation, although direct neurocognitive benefits require further evaluation.

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Introduction

The Anthropocene is characterized by a rapidly intensifying climate crisis, with human exposure to extreme heat events increasing across many regions. A growing body of research has identified a less visible but potentially important consequence: vulnerability of cognitive and mental functioning during heat exposure. This concern extends beyond the well-documented acute physiological effects of excessive heat, such as heat exhaustion, heatstroke, cardiovascular strain, and renal stress. Recent global assessments emphasize that climate-related heat exposure is already affecting population health and labor capacity, making the cognitive and psychological dimensions of heat an increasingly relevant public-health concern [1].

At its core, heat-related cognitive susceptibility reflects the interaction between thermoregulation, cardiovascular strain, hydration, perceived discomfort, and the cognitive demands of the task. Reviews of experimental studies indicate that heat strain can impair some domains of attention, working memory, and executive performance, but findings are heterogeneous and depend on core temperature, exercise, task complexity, acclimatization, and study design [2,3]. Field evidence is consistent with this heterogeneity: during a heat wave, young adults living in non-air-conditioned residences showed slower performance on tests of selective attention, processing speed, and working memory than peers living in air-conditioned residences [4]. Accordingly, the available evidence does not support a single universal Wet-Bulb Globe Temperature (WBGT) threshold at which cognitive failure begins; cognitive effects are better interpreted as task- and exposure-dependent [2,3].

The Global South may carry a disproportionate burden because high ambient temperatures often coincide with labour-intensive occupations and limited access to cooling. South Asia is a major vulnerability hotspot. Recent Sri Lankan occupational evidence from apparel-sector small and medium enterprises found widespread worker-reported heat stress and linked diminished concentration during hot conditions with safety problems, including needle-prick injuries [5]. Although this evidence does not establish a quantified loss of cognitive efficiency, it illustrates how heat-related concentration difficulties may translate into occupational risk in resource-constrained settings.

The socioeconomic consequences of heat-related reductions in health, labor capacity, and productivity are substantial [1,6]. Interventions such as occupational heat protection, access to cooling, heat-health warning systems, and urban heat mitigation therefore have relevance beyond immediate thermal comfort. The present review addresses these intersecting physiological, psychological, and social concerns by synthesizing evidence on the neurocognitive effects of heat stress, proceeding from mechanistic evidence to epidemiological findings, and then to policy implications. The novelty of this review lies in linking evidence across molecular neurobiology, cognition, mental health, vulnerable life stages, occupational exposure, and urban heat mitigation within a single neurocognitive framework, while distinguishing established findings from emerging or indirect evidence.

The Anthropocene Framework: Evolutionary Mismatch, the Polycrisis, and Climate Trauma

The current geological era is widely discussed through the concept of the Anthropocene, which emphasizes the scale of human influence on Earth's systems [7]. At the intersection of human biology and accelerated environmental change, evolutionary mismatch provides one conceptual lens through which to consider whether inherited physiological and behavioral adaptations are well suited to rapidly changing technological and environmental conditions [8]. This framework is useful as an interpretive perspective, although it should not be treated as direct evidence that contemporary heat exposure produces neurocognitive impairment through evolutionary mismatch itself.

In the context of extreme heat, modern built environments can both protect and expose populations. Mechanical cooling reduces immediate heat strain, while urban design, housing quality, power reliability, outdoor work, and the urban heat island (UHI) effect influence the intensity of exposure. Evolutionary-mismatch perspectives have been used to discuss how rapidly changing technological environments may interact with human adaptive systems [8,9], but the heat-health consequences of urbanization are more directly supported by environmental and physiological evidence [10].

This broader environmental disruption also has psychological consequences. Climate-related worry, grief, and anxiety are increasingly documented, particularly among younger people [11]. These experiences overlap conceptually with stress and trauma responses, but current evidence does not justify treating climate anxiety as neurologically equivalent to post-traumatic stress disorder or assigning it a specific pattern of frontal or amygdala dysfunction.

Anticipatory distress about climate change may affect well-being, perceived agency, and future outlook. In a survey of 10,000 young people aged 16-25 years across 10 countries, 59% reported being very or extremely worried about climate change [11]. The study also documented negative emotions and perceived government inadequacy. Such findings indicate a substantial psychological burden, while also showing why climate-related distress should be distinguished from a formal psychiatric diagnosis.

Cellular and Molecular Pathways of Heat-Induced Neurotoxicity

The molecular mechanisms linking severe heat exposure to neurological dysfunction involve oxidative stress, endothelial injury, blood-brain barrier (BBB) disruption, systemic inflammation, and neuroinflammatory responses. Heatstroke represents the most severe clinical end of this spectrum and can lead to multi-organ dysfunction and neurological injury [12]. Experimental reviews of heat strain also show that cognitive effects become more likely as physiological strain increases, although there is no single core-temperature value that predicts impairment across all cognitive tasks [2].

Transcriptomic investigations of heatstroke models have provided insight into changes in the neuroimmune microenvironment after intense heat exposure [13]. These studies identify alterations in inflammatory signaling and immune-cell profiles, while broader heatstroke literature supports disruption of endothelial and BBB integrity during severe hyperthermia [12,13]. Because much of this mechanistic evidence comes from animal or severe-illness models, direct extrapolation to ordinary community heat exposure requires caution.

Gene expression and cellular deconvolution analyses after experimental heatstroke show increased inflammatory cell signatures and changes in markers associated with activated immune responses. Importantly, these findings indicate a mixed and dynamic neuroimmune response rather than a simple conversion of astrocytes or microglia from a single 'protective' state to a single 'neurotoxic' state [13].

Heatstroke-associated inflammation can involve cytokine release, oxidative stress, endothelial injury, and matrix remodeling processes that may contribute to BBB dysfunction, cerebral edema, and neuronal injury [12,13]. These mechanisms are biologically plausible contributors to acute neurological complications of heatstroke, but their contribution to milder ambient-heat-related cognitive changes in humans remains less certain.

Neuroinflammation is also central to contemporary models of neurodegenerative disease, including interactions among activated microglia, astrocytes, oxidative stress, amyloid processing, and tau pathology [14]. This literature provides biological plausibility for studying environmental heat as a potential modifier of neuroinflammatory burden, but it does not establish that ordinary heat exposure causes Alzheimer's disease, Parkinson's disease, or amyotrophic lateral sclerosis.

Molecular studies of severe heat injury have identified broad changes in inflammatory and neural tissue markers [12,13]. However, such profiles are best interpreted as indicators of acute heat injury and neuroimmune disturbance rather than as evidence that heat exposure

reproduces an end-stage neurodegenerative disease phenotype.

Emerging evidence also implicates the gut-liver-brain axis in heat-related neurological effects. In a murine model of periodic heat-wave exposure, transcriptomic profiling identified coordinated changes across the gut, liver, and brain and highlighted orosomucoid 2 (ORM2) as one potential mediator within a liver-brain inflammatory pathway [15]. These findings are hypothesis-generating and require confirmation in human studies before specific causal pathways can be established.

Sleep Architecture Disruption as a Mediator of Cognitive Impairment

Beyond direct cellular injury, high nighttime temperatures may influence cognition indirectly through sleep disruption. Thermoregulation and sleep are closely coupled, and controlled physiological research shows that thermal conditions can alter sleep architecture [16]. Population-level evidence also links warmer nights with insufficient sleep, while global wearable-device data indicate that rising nighttime temperatures shorten sleep, with larger effects among older adults and residents of lower-income countries [17,18]. The cognitive consequences of heat-related sleep loss are, nevertheless, not yet quantified consistently across populations.

Before sleep onset, peripheral vasodilation contributes to the normal decline in core body temperature. Experimental work demonstrates that relatively small changes in the thermal environment can alter slow-wave sleep in healthy adults [16]. These findings support a thermoregulatory link between environmental temperature and sleep architecture, while also showing that responses depend on the timing and pattern of temperature exposure.

In a small controlled study of healthy young men, gradual changes in room temperature that lowered and delayed the minimum core body temperature altered slow-wave sleep without producing a simple uniform deterioration in total sleep time or sleep efficiency [16]. Consequently, the literature should not be interpreted as showing that every 2 °C reduction necessarily improves sleep or that a specific amount of nocturnal heat produces predictable deficits in memory, intelligence, or mathematical reasoning.

The sleep pathway remains important because disrupted sleep is independently associated with poorer attention, memory, and emotional regulation. However, direct longitudinal evidence linking heat-related sleep disruption to later cognitive impairment remains limited. Studies that simultaneously measure ambient heat, objective sleep, and repeated cognitive outcomes are needed to clarify mediation and temporal sequence.

Persistent nocturnal heat may also compound discomfort, irritability, and psychological strain. Sleep loss itself can worsen mood and emotional regulation, but in the context of extreme heat it is only one of several plausible pathways; physiological strain, medication effects, social conditions, pre-existing illness, and access to cooling may also contribute [17,18].

Prenatal and Pediatric Neurodevelopmental Vulnerability

Prenatal and pediatric populations may be particularly vulnerable to climate-related heat exposure because thermoregulation, organ development, and dependence on caregivers differ from those of healthy adults. Pediatric climate health guidance identifies children as a population with distinctive climate-related health vulnerabilities [19]. Nevertheless, developmental vulnerability should not be interpreted as proof that all heat exposures produce neurological injury.

The first 1,000 days of life, from conception through early childhood, represent a sensitive developmental period. A recent scoping review of prenatal ambient heat and neurodevelopment found stronger mechanistic evidence in animal studies than in human studies and concluded that the human evidence remains limited and heterogeneous [20]. This supports concern about possible effects of maternal heat exposure while also requiring caution about strong causal claims concerning congenital or long-term neurodevelopmental abnormalities.

Neuroimaging research is beginning to explore whether prenatal heat modifies brain development. A study of children exposed

prenatally to Superstorm Sandy reported that extreme ambient heat may amplify associations between disaster exposure and basal-ganglia volume, including regions such as the putamen, caudate, and pallidum [21]. Because this is an observational study involving combined exposures, the findings should be interpreted as an association rather than as evidence that heat alone caused the structural differences.

Evidence linking prenatal heat exposure with specific neurodevelopmental diagnoses remains emerging. Recent human studies have reported associations with developmental delay and, separately, with cerebral palsy under certain exposure definitions [22,23]. However, the available evidence does not support a claim that each 1 °C increase in temperature variance produces a 3% increase in the risk of cerebral palsy, language delay, and neurodevelopmental delay, nor does it establish heat as a cause of autism spectrum disorder or attention-deficit/hyperactivity disorder.

A large birth-cohort study of 67,453 child-mother pairs in Foshan, China, used temperature data to examine maternal heatwave exposure during pregnancy and found that exposure during early and late pregnancy was associated with higher risks of neurodevelopmental delay across assessed domains [22]. The authors also emphasized that evidence in this field remains limited, and the observational design does not establish causality.

Co-occurring heat and air pollution may create additional developmental risks, although these exposures should be analyzed separately rather than assumed to share a single mechanism. A systematic review and meta-analysis found an association between childhood fine particulate matter (PM2.5) exposure and lower intelligence quotient (IQ) scores [24]. Evidence on prenatal heat and neurodevelopment remains less mature [20,22]; thus, combined-exposure mechanisms require further study.

Psychiatric Morbidity, Severe Mental Illness, and Climate Trauma

The association between elevated ambient temperatures and psychiatric outcomes is a rapidly developing area of global public health. A systematic review and meta-analysis found that higher temperatures and heatwaves were associated with increased risks of several adverse mental health outcomes, including mental-health-related morbidity and suicide [25]. Large-scale US healthcare data also show that days of extreme heat are associated with higher rates of emergency department visits for mental health conditions, strengthening the evidence that temperature extremes have clinically relevant population-level effects [26].

Meta-analytic evidence indicates that suicide risk increases with higher ambient temperatures, although estimates vary by exposure metric and setting. The meta-analysis reported increases in suicide risk with both daily and monthly temperature increments [25]. Earlier modeling from the United States and Mexico projected approximately 9,000-40,000 additional suicides by 2050 under different warming scenarios rather than a single fixed estimate of 22,000 [27]. Proposed neurochemical and behavioral mechanisms remain plausible but are not sufficient to establish a single causal pathway from heat to suicide.

Individuals with pre-existing severe mental illness (SMI), including schizophrenia-spectrum and mood disorders, can be particularly vulnerable during extreme heat episodes [10]. Risks may reflect a combination of illness-related factors, social vulnerability, comorbid physical disease, limited access to cooling, and medication effects. Some psychotropic and anticholinergic medicines can interfere with thermoregulation or sweating, which warrants clinical attention during heat events [10].

Even among individuals without a psychiatric history, extreme heat is associated at population level with psychological distress and adverse mental health outcomes [25]. Evidence differs across locations, diagnoses, and study designs, and not every reported association can be assumed to reflect a direct neurobiological effect of temperature. Large-scale emergency department data strengthen the evidence for acute service use effects, but observational designs do not by themselves establish a direct neurobiological mechanism [26].

Intersectional Disparities: Demographic, Socioeconomic, and Occupational Vulnerabilities

The neurocognitive and physiological effects of extreme heat are not distributed uniformly. Age, socioeconomic resources, neighborhood conditions, occupational exposure, comorbidity, and access to cooling can modify risk. Longitudinal evidence from older adults in the United States suggests that cumulative exposure to extreme heat is associated with faster cognitive decline in some groups [28].

In a longitudinal US study of older adults, high cumulative extreme heat exposure was associated with faster cognitive decline among Black participants and residents of disadvantaged neighborhoods, but not among White or Hispanic participants or residents of wealthier neighborhoods [28]. These findings indicate effect modification by social context and caution against relying only on population-average estimates.

Such disparities are consistent with broader environmental health concerns about neighborhood disadvantage, housing, urban heat islands, tree canopy, and access to cooling. The cohort findings support the interpretation that social and neighborhood conditions can modify the cognitive consequences of repeated heat exposure [28]. However, the study remains observational and cannot isolate individual mechanisms such as housing quality, air-conditioning access, or lifetime structural disadvantage.

Occupational and outdoor heat exposure represents another intersection of cognitive performance and environmental vulnerability. A quasi-experimental field study in Singapore examined 309 older adults during outdoor activity and assessed thermal comfort, environmental conditions, cognitive control, and pro-environmental intentions. The findings linked poorer outdoor thermal comfort with reduced cognitive control and showed that cognitive control was relevant to the translation of environmental attitudes into behavioral intentions [29].

This Singapore study provides useful field evidence because it combines outdoor thermal conditions with a cognitive control measure. The study did not involve occupational workers or 24 worksites; rather, it included older adults participating in outdoor activities. Its findings therefore support an association between thermal comfort and cognitive control rather than a general estimate of workplace heat exposure [29].

In a low-resource setting, ethnographic research from Burkina Faso documented how extreme heat affected pregnant women's physical comfort, daily activities, sleep, well-being, and ability to cope with pregnancy-related demands [30]. The qualitative findings highlight the interaction between heat exposure, social conditions, and maternal well-being without implying a quantified neurocognitive deficit.

Discussion

The evidence synthesized in this review supports a clinically important but more nuanced conclusion: extreme heat can adversely affect neurological, cognitive, and mental health outcomes through multiple pathways, but the strength of evidence differs by outcome and exposure severity (Figure 1). Evidence is strongest for acute heat illness, population-level mental health associations, and task-dependent cognitive effects during heat strain; mechanistic, prenatal, and long-term neurocognitive pathways remain important but in several areas are still emerging.

Despite growing evidence that heat affects both physical and mental health, heat-health planning has historically prioritized physical morbidity and mortality. A 2025 synthesis of heat action plans from six countries identified substantial variation in governance, implementation, vulnerability assessment, multi-sectoral coordination, and evaluation [31]. The report highlights the importance of adaptable core elements, clear agency roles, early warning, and integration of short-term response with long-term planning.

The available policy evidence supports stronger integration of mental health risks, medication-related vulnerability, communication, cooling access, and continuity of care within heat-health planning, particularly for people with pre-existing illness [10,25].

People with pre-existing mental illness are an important vulnerable group during extreme heat because clinical, pharmacological, social, and environmental factors may interact. Heat

health plans should therefore avoid treating vulnerability as a single age-based category and should consider medication review, hydration, social support, and access to cooling where clinically appropriate [10].

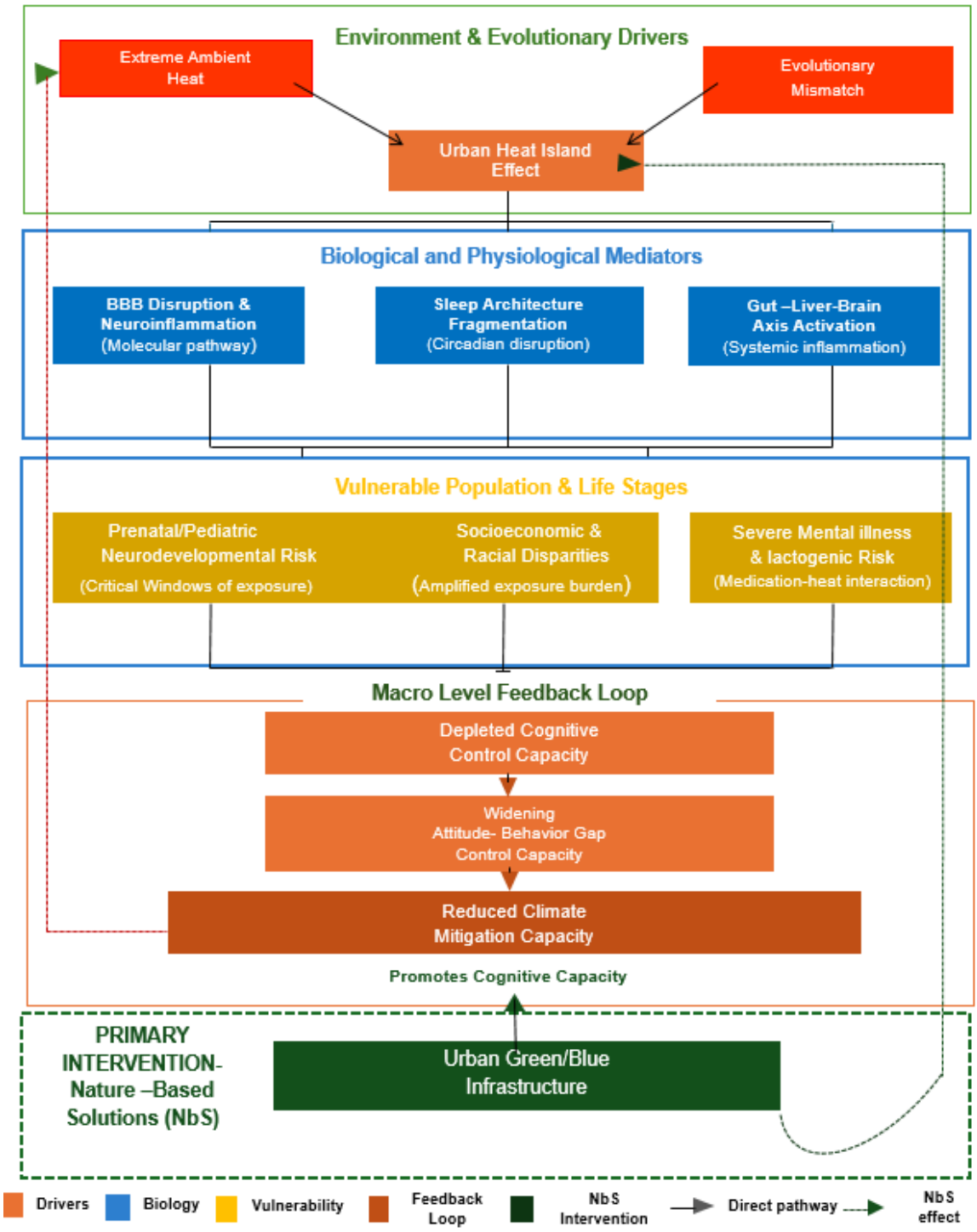


Figure 1: Multiscalar framework of heat-induced neurocognitive vulnerability

Note: Solid arrows (→) indicate proposed or evidence-supported pathways from environmental stressors to molecular, sleep, and life-stage effects; they should not be interpreted as proof of causality for every pathway. Dashed arrows indicate potential mitigating and restorative effects of NbS. The feedback arc represents a conceptual hypothesis linking impaired cognitive control with reduced capacity for pro-environmental action.

Geographic inequity remains a central policy concern because heat exposure, health system capacity, housing quality, occupational risk, and access to cooling vary substantially between and within countries [1,10]. Because the available heat action plan synthesis evaluated six countries, generalization to all national and local heat planning systems should remain cautious.

Global policy frameworks should move from general awareness toward practical, multisectoral heat-health action. Priorities include early warning, occupational protection, accessible cooling, continuity of medical and mental healthcare, and review of heat-sensitive medications for high-risk patients [10,25]. The appropriate intervention mix should be adapted to local climate, health system capacity, and population vulnerability rather than treated as a single mandatory protocol.

NbS as Heat Mitigation and Potential Neurocognitive Interventions

Addressing heat-related neurocognitive vulnerability requires both clinical adaptation and modification of the environments in which people are exposed. NbS, including urban trees, green corridors, and other forms of green and blue infrastructure, can reduce urban heat through shading and evapotranspiration [32]. These approaches may also provide broader environmental and well-being benefits, although direct evidence that NbS independently restores neurocognitive function remains limited.

Urban greening can produce meaningful local cooling, but the magnitude varies by climate, design, vegetation, scale, and measurement method [32,33]. In Medellín, Colombia, the Green Corridors initiative converted 18 roads and 12 waterways into greener corridors and was reported to have reduced local temperatures by more than 2 °C. Urban nature initiatives in cities including Melbourne and Mumbai have also been described as components of wider climate adaptation strategies. These examples demonstrate implementation potential but should not be interpreted as controlled evidence of cognitive recovery. Complementing these implementation examples, a recent study from Colombo District, Sri Lanka, reported that emotional restoration and cognitive-reflective engagement with natural environments were independently associated with life satisfaction, providing locally relevant but preliminary evidence for psychological co-benefits of nature engagement [34].

Structural barriers to widespread NbS adoption include limited space, financing, maintenance, governance, and the need for locally appropriate evidence. Evidence from Asia indicates that cooling performance is context-dependent and influenced by the type, size, geometry, distribution, climate, and assessment method of green and blue infrastructure [32]. Earlier systematic evidence similarly supports local cooling effects while emphasizing substantial heterogeneity and limitations in intervention evidence [33]. Green infrastructure is therefore best framed as a heat mitigation and public health strategy with potential neurocognitive co-benefits, rather than as established 'neurological protective equipment.'

The Cognitive-Behavioral Feedback Loop in Climate Mitigation

A further hypothesis arising from this review concerns whether heat-related changes in cognitive control could influence pro-environmental decision-making. Environmental attitudes do not always translate into behavior, and field evidence from Singapore suggests that thermal comfort and cognitive control may help explain part of this gap [29].

The Singapore study found that outdoor thermal comfort was associated with cognitive control, which in turn was relevant to pro-environmental behavioral intentions [29]. This provides an empirical basis for considering cognition within climate-behavior research, but it does not demonstrate that heat exposure at the population level directly prevents people from undertaking specific actions such as recycling, public transport use, or energy conservation.

Accordingly, the proposed cognitive behavioral feedback loop should be understood as a conceptual extension of the available evidence rather than an established causal cycle. If heat-related discomfort or cognitive strain reduces cognitive control, it is plausible that complex

goal-directed behaviors could become more difficult; however, population-level consequences for climate mitigation require direct testing [29]. Policies that protect thermal health may therefore have behavioral co-benefits, but these should be evaluated empirically.

Limitations

Several important limitations must be acknowledged. First, as a narrative review, this synthesis was not based on a pre-registered protocol, exhaustive systematic screening, or formal risk-of-bias assessment and is therefore susceptible to selection bias. Second, evidence differs substantially in design and strength: some mechanistic findings derive from animal or severe heatstroke models, whereas many human studies are observational. Third, several areas, particularly prenatal neurodevelopment, long-term cognitive decline, and cognition-mediated climate behavior, remain emerging and should not be interpreted causally. Fourth, high-resolution neurocognitive evidence remains geographically uneven, limiting generalizability to low-resource and rural settings. Fifth, heat health policy is evolving rapidly, so current planning documents may change after publication. Future research should prioritize longitudinal, multi-site studies with objective heat exposure, sleep, physiological, and neurocognitive measures and stronger causal inference designs.

Conclusion

This narrative review synthesizes evidence from physiology, neurobiology, epidemiology, psychiatry, developmental health, occupational research, and urban planning to show that extreme heat is an important neurocognitive and mental health concern in the Anthropocene. The evidence is strongest for acute heat illness, task-dependent cognitive effects during heat strain, and population-level associations with adverse mental health outcomes. BBB disruption and neuroinflammatory cascades are well supported in severe heatstroke and experimental models, whereas prenatal neurodevelopmental effects, long-term cognitive decline, and cognition-mediated climate-behavior pathways remain emerging. Vulnerability is shaped by age, health status, occupation, socioeconomic conditions, and access to cooling. Heat health policy should therefore incorporate mental health and medication-related vulnerability alongside established physical health protections. Urban greening and other NbS can contribute to heat mitigation, but their direct neurocognitive benefits require further evaluation. Protecting cognitive and mental health should be considered an important component of climate adaptation without overstating causal mechanisms that have not yet been established.

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